



CHE372 Fall 2026 Syllabus

Course Number: CHE 372

Course Title: Reactor Design

Unique Number: 17155

⚠ Changes to the schedule and syllabus may be made at the instructor's discretion. It is your responsibility to note these changes when announced.

Personnel

Instructor: Dr. Benjamin K. Keitz

Teaching Assistant: Meng-Hsiu Shih

Tutor: Zachary Hammer

Course Goals and Overview

Lectures

Recitation

Grading and Major Assignments

Exams

Homework

Homework Quizzes

Lecture Quizzes

Project

Regrade Requests

Late Work

[Absences](#)

[Student Outcomes and ABET Program Outcomes Achieved](#)

[Knowledge, Abilities, and Skills Students Should Gain From This Course](#)

[Impact on Subsequent Courses in the Curriculum](#)

[Course Prerequisites](#)

[Course Textbooks](#)

[Additional Resource Requirements](#)

[Course Website](#)

[Approximate Course Outline \(subject to change\)](#)

[Academic Integrity](#)

[Q Drop Policy](#)

[Sharing of Course Materials is Prohibited](#)

[Use of Generative AI](#)

[Emergency Preparedness and Emergency Plan Instructions](#)

[Americans with Disabilities Act](#)

[Title IX Reporting](#)

Personnel

Instructor: Dr. Benjamin K. Keitz

keitz@utexas.edu

Office Hours: TBA

Teaching Assistant: Meng-Hsiu Shih

menghsiushih@utexas.edu

Office Hours: TBA

Tutor: Zachary Hammer

zsh354@my.utexas.edu
Office Hours: TBA

Course Goals and Overview

Chemical Reactor Analysis and Design is a unique, capstone course in the chemical engineering curriculum that distinguishes our field from other engineering disciplines. This course can be seen as a culmination of your undergraduate education in chemistry, material and energy balances, transport phenomena, numerical methods, and thermodynamics. Overall, this course will provide an introduction to the principles and calculation techniques used to analyze and design chemical reactors for the production of petrochemicals, advanced materials, polymers, biochemicals, and biologics. This course is designed to:

- Provide you with the fundamentals of material and energy balances as applied to chemical reactor design for ideal reactors
- Offer practice in defining problems, analyzing data, and designing chemical processes.
- Provide you with an understanding of rate laws and their derivation.
- Provide you with an appreciation for the topics of heterogeneous catalysis, biological catalysis, and non-ideal reactors

Lectures

TTh (Tuesdays and Thursdays) : 12:30 pm - 2 pm : **CPE 2.212**

Please be punctual: In person attendance and class participation are strongly encouraged. This class will be a partially flipped classroom where students review lecture material before class and lectures are spent going through example problems. Lectures will be recorded for later upload and can be accessed using the Panopto link on Canvas. Written notes and completed example problems will also be uploaded to Canvas.

Recitation

W (Wednesdays) : 8 am - 9 am : **CPE 2.216**

Attendance is not mandatory at recitation but is strongly suggested since the TA will be going over practice problems, coding tutorials, answering questions, and otherwise supplementing lecture materials. If you need additional help with coding, attending recitation is highly encouraged.

Grading and Major Assignments

Course grades will be assigned based on the following grading rubric:

Exams (best 3 of 4)	60%
Homework	5%
Homework Quizzes	10%
Lecture Quizzes	5%
Project	20%

The following grade cutoffs will be used: $A \geq 93$; $A^- = 93-90$; $B^+ = 90-87$; $B = 87-83$; $B^- = 83-80$; $C^+ = 80-77$; $C = 77-73$; $C^- = 73-70$; $D < 70$; $F < 60$. Grades will be curved as necessary and plus/minus grading will be used for this course.

Exams

Three exams are scheduled on **September 24, 2026**, **October 22, 2026**, and **December 3, 2026**. All exams are in person and will take place from 6-8 pm CDT. There will be no make up exams offered! Please double check for potential conflicts with exams in other classes and let the instructor know as soon as possible. Optional review sessions will be held before each of the exams and will be announced through Canvas. The three mid-term exams will have the following structure:

Exam 1: No resources besides a graphic calculator. A formula sheet will be provided to you.

Exam 2: Single page (front and back) of handwritten notes. A formula sheet will also be provided.

Exam 3: Open notes (digital) and full computer access (including AI).

The Final Exam for the course is optional and is set by the registrar for December XX, 2026 1-3 pm. The location of the final exam will be announced the last week

of class. If you choose to take the final exam, the best 3/4 Exam scores will count towards your final grade.

Homework

The purpose of homework is for you to practice example problems without the instructors assistance. Homework will be assigned Thursdays via Canvas and is due one week after it is assigned (the following Thursday). One problem from the homework will be selected at random for grading, while the other problems will be graded for completion. You are strongly encouraged to work in small groups to discuss and solve the homework problems prior to the due date. All homework should be scanned and submitted through Gradescope. Your lowest homework grade will be dropped.

Homework Quizzes

The purpose of homework quizzes is for you to practice completing problems under exam-like time constraints. A quiz containing a problem similar to one or two questions from the homework will be given the week after it is assigned (the following Thursday, the same day homework is due). Quizzes will be administered remotely through Gradescope and can be completed anytime within a ~12 h window. Each quiz will have a defined time limit, for example, once you start the quiz you must complete the problem and upload your solution within 30 minutes. Quizzes must be completed individually and will require scanning and uploading written solutions. Your lowest quiz grade will be dropped.

Lecture Quizzes

The purpose of lecture quizzes is to evaluate your understanding of previously assigned lecture material. There will be one or two lecture quizzes per week and they will be announced beforehand. Lecture quizzes will be given at the beginning of class and consist of multiple choice questions. UT Instapoll will be used for lecture quizzes. Your lowest two lecture quiz scores will be dropped.

Project

The purpose of the project is to demonstrate how the concepts learned in the class apply to the many subfields of Chemical Engineering (process control, genetics, biology, petrochemicals, polymers, etc.). The project primarily consists

of a written report. Specific instructions and a written template will be provided later in the semester. You will be randomly assigned to a team of 3-4 students and given a topic to work on. More detail about the project topics and teams will be provided later in the semester.

Regrade Requests

You have one week after an assignment is returned to you to request a re-grade. Re-grading requests should be submitted in writing at the end of the lecture period and should consist of a few sentences on a *separate* piece of paper that describe your rationale for your request. Nothing is to be marked on the original assignment or else the re-grade request will be voided. Re-grade requests may also be submitted using Gradescope.

Late Work

Assignments must be turned in by the assigned due date. Late work will have 20% deducted for every 24 hours past the original due date. For example, if an assignment is due Thursday at midnight, submitting it Friday at 1 am would result in 20% off, submitting it Saturday at 1 am would result in 40% off, etc. Assignments turned in more than 48 hours past the original due date will not be accepted.

Absences

There are no excuses for missed exams or quizzes other than written certified medical excuses or written letters on university letterhead for UT-related school activities. By UT Austin policy, you must notify the instructor of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holiday, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Student Outcomes and ABET Program Outcomes Achieved

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- An ability to design a system, component, or process to meet desired needs within realistic constraints such as economical, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.
- An ability to identify, formulate, and solve engineering problems.
- An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

Knowledge, Abilities, and Skills Students Should Gain From This Course

- Ability to derive and apply the energy and material balances that are required to design isothermal and non-isothermal batch, plug flow, fixed-bed, and continuous stirred tank reactors.
- Ability to solve problems of variable density and multiple independent reactions.
- Ability to solve problems of mass transfer with reactions in heterogeneous catalysts.
- Ability to solve problems incorporating simple, one-parameter non-ideal reactor modeling.
- Knowledge of chemical reaction analysis concepts and chemical kinetics concepts.

Impact on Subsequent Courses in the Curriculum

Virtually every subsequent CHE course requires material balance concepts as well as computational skills.

Course Prerequisites

- A grade of at least C in CHE 322, CHE 348, and CHE 354.
- Knowledge of heat transfer processes (CHE 354 and CHE 353)

- Understanding of reaction equilibria and heat of reaction (CHE 322)
- Ability to solve linear differential equations (M 427K)
- Familiarity with matrix multiplication (CHE 348)
- Ability to solve ODEs using explicit numerical methods and to have a working knowledge of how these methods work (CHE 348)
- Ability to solve sets of linear and nonlinear algebraic equations numerically (CHE 348)
- Familiarity with numerical optimization procedures (CHE 348)
- Working knowledge of MATLAB, Mathematica, Python, Julia, or equivalent software for solving differential and algebraic equations and plotting results (CHE 348).
- A comprehensive knowledge of transport phenomena (CHE 319)

Course Textbooks

You are not required to purchase a textbook, but the textbooks listed below may have example problems and additional discussion that you may find helpful.

- *Chemical Reactor Analysis and Design Fundamentals* by Rawlings and Ekerdt
- *Elements of Chemical Reaction Engineering* by Fogler
- *Chemical Reaction Engineering* by Levenspiel
- *Chemical Reactor Analysis and Design* by Froment and Bischoff

An electronic copy of *Fundamentals of Chemical Reaction Engineering* by Davis and Davis will also be made available on the course website.

Additional Resource Requirements

You are required to have the following:

- Laptop computer with Microsoft Office as per the latest [UT Engineering](#) standards.
- Software for obtaining numerical solutions for partial differential equations. We will mostly use Python in this course.

- Several class resources to supplement lecture material will be released as Python code / Jupyter notebooks. The easiest way to run and visualize these examples is using Anaconda to run a Python environment on your personal computer.
- There will be several recitations devoted to learning the basics of Python, going through examples, and troubleshooting Python installations.

Course Website

Please check the course webpage on Canvas regularly. All announcements will be provided through Canvas.

Approximate Course Outline (subject to change)

Introduction to Chemical Reactors and Material Balances

Batch Reactors

Continuous Stirred Tank Reactors (CSTRs)

Semi-Batch Reactors

Plug Flow Reactors

Multiple Reactions and Reactor Selection

Defining Rate Laws

Overall Energy Balance for Reactors

Batch Reactors

CSTRs

PFRs

Overview of Heterogeneous Catalysis

Defining Catalysis Rate Laws

External Diffusion Considerations

Internal Diffusion Considerations

Packed Bed Reactor Design

Energy Balance Considerations

Enzyme Reactions

Bioreactors

Pharmacokinetics
Residence Time Distributions and Real Reactors
Data Analysis

Academic Integrity

The University has a strict policy on academic integrity. Any form of plagiarism or academic dishonesty will NOT be tolerated in CHE 372. If you have any questions, please consult the following websites:

Students are obliged to follow the Student Honor Code that states "As a student of the University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity."

Failure to comply with the University's policy will result in a "zero" on the assignment and an "F" in CHE 372.

Q Drop Policy

If you want to drop the course after the 12th class day, you will need to execute a Q drop before the Q-drop deadline, which typically occurs near the middle of the semester. Under Texas law, you are only allowed six Q drops while you are in college at any public Texas institution. For more information, please see:

Sharing of Course Materials is Prohibited

No materials used in the class, including, but not limited to lecture hand-outs, videos, assessments (quizzes, exams, etc.), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have the instructor's express written permission.

Sharing course materials for the purpose of incorporating them into machine learning models is also prohibited.

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Use of Generative AI

Students are invited to use AI/LLM platforms including ChatGPT, Claude, Gemini, etc. to help prepare for assignments and projects. Unless explicitly stated otherwise, you are welcome to use AI tools to help *revise* and *edit* your work. In all cases, you should start with an original draft. When submitting work, students should clearly identify any writing, text, or media generated by AI. This can be done in a variety of ways. You are also encouraged to use generative AI platforms for developing Python, MATLAB, or Julia code. This can be particularly helpful if you are struggling with syntax or encountering bugs in your code. If you are relying on these services, please note that they usually web-based (not local) and may not be available during an exam.

Emergency Preparedness and Emergency Plan Instructions

Please take note of all emergency exits, exit procedures, and University Emergency Communications Policies for the lecture and recitation buildings. Students requiring assistance in evacuation should inform the Professor of the course in writing of their needs during the first week of class so that this information is properly transmitted to the Fire Prevention Services office.

Americans with Disabilities Act

The University of Texas at Austin is on record as being committed to both the spirit and letter of federal equal opportunity legislation; reference Public Law 92-112 - The Rehabilitation Act of 1973 as amended. With the passage of federal legislation entitled Americans with Disabilities Act (ADA), pursuant to section 504 of the Rehabilitation Act, there is renewed focus on providing this population with the same opportunities enjoyed by all citizens.

As a faculty member, I am required by law to provide “reasonable accommodations” to students with disabilities, so as not to discriminate on the basis of that disability. Student responsibility primarily rests with informing the instructor of their need for accommodation and in providing authorized documentation through designated administrative channels. Information regarding

specific diagnostic criteria and policies for obtaining academic accommodations can be found at the following website:

Students with accommodations must register for alternative exam times/lengths through the Student Testing Center:

Title IX Reporting

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating. 2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation. 3.

Investigate and discipline violations of the university's relevant policies(<https://titleix.utexas.edu/relevant-policies/>). Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee.

If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as sexual misconduct, including the types of sexual misconduct covered under Texas Senate Bill 212.

The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.